



Natural language processing course project report

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Abstract

We built a RAG-based assistant for the UL FRI student office that answers administrative queries about study programmes and procedures in Slovenian and English. The pipeline crawls and parses UL/FRI web content, splits it into 400-token section-aware chunks, embeds them with multilingual sentence-transformers, and retrieves candidates via hybrid BM25+dense search followed by cross-encoder reranking. We compare three embedders and three retrieval modes on 51 in-scope and 10 out-of-corpus questions using document-level metrics (Recall@ k , nDCG@ k) and a RAGAS-style LLM judge scoring faithfulness, hallucination, and refusal quality. The cross-encoder reranker is the single largest contributor, raising in-scope faithfulness from 0.575 (dense baseline) to a reference ceiling of 0.748; hybrid retrieval alone adds little until reranking is applied. Human evaluation selects `bge-m3` with hybrid+rerank ($k=6$, $k_r=30$) as the best RAG configuration at 5.1/10, with `Llama-3.1-8B-Instruct` achieving 0.665 LLM-judged faithfulness and 7/10 correct out-of-corpus refusals. The gap between the best RAG system and direct Claude calls (8.6/10) identifies retrieval quality as the primary remaining bottleneck.

Keywords

Retrieval augmented generation, dense retrieval, hybrid retrieval, multilingual embeddings, student affairs chatbot

Introduction

We built an assistant for the student office of the Faculty of Computer and Information Science (UL FRI) to answer everyday administrative questions about study programmes, regulations, and office procedures in Slovenian and English. Large language models alone are a poor fit for this task: answers often require precise details (deadlines, paragraph numbers, application forms) that must come from official documents rather than the model's training data. For this reason we used Retrieval Augmented Generation [1], which has proven useful for university-style FAQ chatbots [2, 3] and helps reduce hallucinations and stale knowledge in educational deployments [4]. The pipeline follows the comparative-retrieval setup (dense vs. hybrid vs. reranked, with a chunk-size sweep) used in prior RAG-for-admissions studies [5, 6], and we evaluate locally hosted instruction-tuned LLMs following work on open-day chatbots and RAG vs. fine-tuning for knowledge-heavy tasks [7, 8].

Methods

Figure 1 gives an end-to-end overview of the pipeline; each stage is detailed below.

Data collection and parsing

The pipeline is organised into *runs*, each run contains raw downloads, parsed JSONL, chunks, and a FAISS index in a single directory. A lightweight crawler walks breadth-first from a curated seed list with separate depth limits for `fri.uni-lj.si`, other `*.uni-lj.si` domains, and external domains. It downloads HTML and linked PDF/DOCX, augments results with sitemaps and RSS/Atom feeds, respects `robots.txt`, deduplicates by SHA-256, and reuses `ETag/If-Modified-Since` headers on updates. Each document is stored as a JSON record (URL-derived `doc_id`, title, language, ordered sections): HTML via BeautifulSoup + `trafilatura`, PDF via `Docling`+`Tesseract` or `PyMuPDF`, and DOCX via `python-docx`.

Chunking, embeddings and retrieval

We split documents with a section-aware sliding window measured in whitespace tokens (default 400, overlap 80). Every chunk is prefixed with the document title and section name, gets a stable `chunk_id`, and is then deduplicated with a SimHash filter. For retrieval experiments we use multilingual sentence-transformer embedders (`intfloat/multilingual-e5-base`, `intfloat/multilingual-e5-large`, and

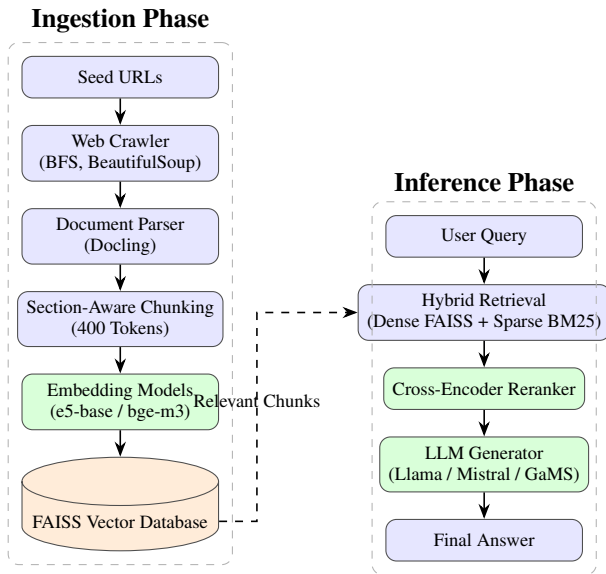


Figure 1. End-to-end RAG pipeline. **Ingestion Phase** (left): web content is crawled breadth-first from seed URLs, parsed with Docling (with PyMuPDF/Tesseract fallback), split into 400-token section-aware chunks (80-token overlap), embedded with a multilingual sentence-transformer, and stored in a FAISS IndexFlatIP. **Inference Phase** (right): at query time, a hybrid retrieval step fuses dense FAISS scores with BM25 scores (min-max normalised), a cross-encoder re-ranks the candidate pool, and the top- k chunks are passed as numbered context to an instruction-tuned LLM.

BAAI/bge-m3) and index chunks in FAISS as an IndexFlatIP (cosine under normalisation). At query time we support three retrieval modes: dense (top- k from FAISS), hybrid (BM25 from `rank_bm25` with default $k_1=1.5$, $b=0.75$, fused with dense scores after min-max normalisation on the union of both top lists), and hybrid plus cross-encoder rerank using `mmarco-mmMiniLmV2-L12-H384-v1` on a candidate pool of size 20. Retrieved chunks are then formatted into a numbered context block and a language-specific system prompt tells the generator to answer *only* from that block and in the detected language.

Evaluation set and reference audit

Our evaluation set is a hand-written JSONL file of 61 questions: 51 *in-scope* (the corpus contains a factual answer) and 10 *negatives* (deliberate out-of-corpus probes used to test refusal), shipped with the repository. Each entry carries the expected language, a list of expected keywords, and a free-form `reference_answer` that we use as the human-written gold for the LLM-as-judge protocol below. A smaller development corpus of 25 questions also carries `relevant_doc_ids` (manually annotated) which we use for the document-level retrieval metrics.

The first time we ran the judge, in-scope faithfulness was very low. Inspection showed many reference answers included details not present in the corpus, which caused the judge to

mark them down. To prevent this reference drift from affecting later comparisons, we performed a Sonnet-graded fact-check before the configuration sweeps: for each positive question we retrieved a top-30 BM25 $\cup$ dense pool from the current index and asked Claude Sonnet (`claude-sonnet-4-6`, $T=0$) to label every claim in the reference as supported, partial, or unsupported, and to optionally produce a grounded rewrite. The application policy was fixed: supported retained the original, partial or unsupported with a rewrite used the rewrite (flagged `auto-rewritten`), unsupported without a rewrite was dropped, and negatives were never modified. The cleaned set was written as a versioned artefact `questions_full.v2.json`.

LLM-as-judge protocol

We used a RAGAS-style judge: for each row we sent the question, the retrieved top- k chunks, the model answer, and the reference answer to Claude with a strict-JSON rubric covering *faithfulness*, *answer relevance*, and *context relevance* (each in $[0, 1]$), a binary *hallucination* flag (set when at least one claim was unsupported by the retrieved context), a *refusal* flag, and an integer *overall* score in $[1, 5]$. Negatives were scored under the same rubric, with a correct refusal rewarded with high faithfulness. All judge calls ran at $T=0$ for reproducibility. To contain costs while maintaining quality at the bottleneck, we used Sonnet (`claude-sonnet-4-6`) for the reference audit and final re-judgment, and Haiku (`claude-haiku-4-5`) for the comparison sweeps. The two models agreed to within 0.01 in in-scope faithfulness on the winning configuration (0.739 vs. 0.748); we therefore used Haiku for ranking.

Results

Corpus, evaluation set, and reference audit

At the default 400/80 chunk policy the full crawl yields 2,742 parsed documents and 29,296 chunks, up from the 176 documents in the initial development corpus. Full Docling OCR takes ~ 4 hours; a faster two-pass pipeline (PyMuPDF + Tesseract fallback) was used for most experiments, with the full OCR pass reserved for the final tests and the distributed FAISS index. Running the Sonnet audit on the original references against this corpus showed that only 9/47 ($\sim 19\%$) of the in-scope references were fully supported as written. Of the rest, 31 were classified *partial* and rewritten using only chunk-attested facts, 7 were *unsupported* (3 dropped, 4 rewritten where Sonnet found alternative chunks), and 14 were skipped (no factual reference, or negative). The cleaned `questions_full.v2.json` contains 51 in-scope (of which 44 have a written reference answer) and 10 negatives, and is the basis for everything that follows.

Document-level retrieval (development corpus)

On the smaller 25-question development corpus, where we have document-level annotations, dense retrieval at $k=4$ already reaches 88.0% Recall@4 and 66.3% nDCG@4. Adding

the cross-encoder reranker lifts these to 100.0% and 93.4%, while hybrid + rerank is essentially tied (100.0% / 93.9%). The corresponding chunk-size sweep on the same corpus showed sizes 200–500 all reaching $\sim 97.9\%$ Recall, with 300 giving the highest precision. The LLM-judged sweep on the full crawl, reported below, revisits this choice on a stricter metric and ends up picking 400.

Embedder $\times$ retrieval-mode sweep (LLM-as-judge)

Table 1 shows the embedder $\times$ retrieval-mode sweep at top- $k=4$ on the cleaned set, judged by Claude Haiku at $T=0$; refusal-side faithfulness was ≥ 0.90 across all six configurations. Two findings are noteworthy. First, the cross-encoder reranker was the largest single contributor: on e5-base it increased faithfulness from 0.575 (dense) and 0.565 (hybrid) to 0.739, and context relevance from about 0.72 to 0.857. Second, e5-base + rerank outperformed e5-large + rerank on every in-scope metric while embedding approximately $2.5\times$ faster, an unexpected result. Once reranking was applied, the smaller first-stage embedder proved sufficient.

Chunk size on the winning configuration

Holding the winner fixed (e5-base, hybrid + rerank, $k=4$) and varying the chunk size with 20% overlap, the Haiku-judged in-scope faithfulness is 0.691 at 200/40 (50,546 chunks), 0.698 at 300/60 (36,331), and 0.739 at 400/80 (29,296). Larger chunks improved performance because the cross-encoder benefited from additional local context per chunk; 400-token chunks also produced the smallest and fastest index of the three, so we selected them as the default.

Reference-grounding ceiling and generator comparison

We re-judged the winning retrieval configuration with Claude Sonnet at $T=0$ in two ways. First, as a *reference-grounding ceiling*: given the retrieved top-4 chunks, how well-supported are the cleaned reference answers themselves? This is an upper bound on what any extractive generator could achieve under our retrieval. The reference set scores 0.748 in-scope faithfulness, 0.826 answer relevance, 0.810 context relevance, 44.2% hallucination, and 10/10 correctly refused negatives, consistent with the Haiku-judged 0.739 and confirming the two judges agree to within 0.01. Walking the gain through the retrieval stages: the original baseline (small corpus, noisy refs, dense $k=8$) sat at Haiku-judged faithfulness of 0.34. The full crawl plus reference audit at dense $k=4$ already moves it to 0.575. Hybrid alone does not help (0.565). Adding the cross-encoder reranker takes it to 0.739 Haiku / 0.748 Sonnet, with per-answer hallucination dropping from 80.9% to 44.2%.

Second, we ran an actual generator comparison on the cluster (onj_fri partition) with the same retrieval, judge, and cleaned set (Table 2). Llama-3.1-8B-Instruct comes out on top at 0.665 in-scope faithfulness, 50.0% hallucination, 7/10 correctly refused negatives. Mistral-7B-Instruct-v0.3 is essentially tied on

faithfulness (0.653) and slightly better on context relevance but slower. GaMS3-12B-Instruct, despite being trained for Slovenian, came out lowest in faithfulness (0.592) with the highest hallucination rate (66.7%) and only 6/10 correct refusals. The gap between the best generator (0.665) and the reference-grounding ceiling (0.748) is small: under our retrieval and judge, most of the remaining headroom is on the retrieval side rather than the generator side.

Human evaluation

As a complementary check we conducted a human 0–10 scoring of system outputs on the full 61-question set (51 in-scope, 10 negative), with a human rater comparing each answer against the reference. All four RAG configurations use Llama-3.1-8B-Instruct as the generator with hybrid retrieval. The three variables are (i) the embedding model (multilingual-e5-base vs. BAAI/bge-m3), (ii) the number of chunks passed to the LLM ($k \in \{4, 6\}$), and (iii) the cross-encoder reranker candidate pool size ($k_r \in \{20, 30, 50\}$). Table 3 shows the results. Scores were revised after auditing the reference answer set: Q10 and Q41 had inverted correct answers, Q43 had transposed phone numbers, and Q13, Q30, Q37 and Q56 lacked references entirely. Claude achieves 8.6/10 mean (revised from 9.1). The best RAG configuration is bge-m3 with $k=6$, $k_r=30$ at 5.1/10, followed by bge-m3 with $k=4$, $k_r=20$ at 4.8/10. Given this result, we revise our system selection to bge-m3 at $k=6$, $k_r=30$ as the final operating setting. Comparing the two default configurations ($k=4$, $k_r=20$), bge-m3 still outperforms e5-base (4.8 vs. 4.2), matching the automated sweep ordering. This evaluation was conducted on 60 questions (Q44 removed as duplicate) and uses a slightly different reference set from the LLM-as-judge sweep. The gap between the best RAG system (5.1) and Claude (8.6) remains substantial, consistent with the automated judge ceiling (0.665 vs. 0.748).

Discussion

Both the retrieval-only and the LLM-as-judge results indicate the same pattern. The largest gains were attributable to two stages: corpus expansion and the Sonnet audit moved dense- $k=4$ faithfulness from 0.34 to 0.575 (about 58% of the total gain), and the cross-encoder reranker increased it further to 0.748 (the remaining 42%). Hybrid retrieval on its own produced negligible improvement; it became beneficial only after reranking was applied, presumably because the union of BM25 and dense candidates provided a more diverse candidate pool for the cross-encoder. Another unexpected finding was that multilingual-e5-base outperformed multilingual-e5-large once reranking was applied while embedding approximately $\sim 2.5\times$ faster, suggesting the cross-encoder absorbed work that would otherwise require a stronger first-stage embedder.

The generator results add another dimension. The best generator (Llama-3.1-8B-Instruct, 0.665 faithfulness) leaves only a 0.083 gap to the reference-grounding ceiling.

Table 1. Embedder $\times$ retrieval-mode sweep on `questions_full_v2.json` (Haiku judge, top- $k=4$, $T=0$). **Faithfulness:** fraction of answer claims supported by the retrieved chunks ($[0, 1]$, higher is better). **Hallucination:** per-answer rate of at least one unsupported claim (lower is better). **Ctx rel.:** judge-rated relevance of retrieved chunks to the question ($[0, 1]$). **Overall:** integer rubric score in $[1, 5]$.

Embedder	Mode	Faithfulness	Hallucination	Ctx rel.	Overall (1–5)
multilingual-e5-base	dense	0.575	59.1%	0.725	3.000
multilingual-e5-base	hybrid	0.565	63.6%	0.715	2.841
multilingual-e5-base	hybrid + rerank	0.739	54.5%	0.857	3.432
multilingual-e5-large	dense	0.680	54.5%	0.812	3.250
multilingual-e5-large	hybrid	0.650	61.4%	0.787	3.205
multilingual-e5-large	hybrid + rerank	0.708	59.1%	0.866	3.364

Table 2. Generator comparison on the cluster (Sonnet judge, $T=0$, same retrieval, top- $k=4$). Same metric definitions as Table 1. n_{is} is the in-scope sample after the judge separates refusals (lower means the generator refused more often). **Refused / 10** counts correctly refused out-of-corpus negatives. The bottom row is the reference-grounding ceiling: Sonnet judging the cleaned references themselves against retrieved chunks; $n=54$ here because 7 in-scope v2 questions added after the audit have no written reference answer.

Generator	n_{is}	Faithfulness	Hallucination	Ctx rel.	Overall	Refused / 10
GaMS3-12B-Instruct	54	0.592	66.7%	0.688	2.907	6/10
Llama-3.1-8B-Instruct	52	0.665	50.0%	0.707	3.000	7/10
Mistral-7B-Instruct-v0.3	51	0.653	58.8%	0.729	2.961	7/10
<i>Reference-grounding ceiling</i>	43	0.748	44.2%	0.810	3.605	10/10

ing (0.748), meaning that under our retrieval and judge, more than half of the remaining headroom is on the retrieval side rather than the generator side. GaMS3-12B-Instruct being the weakest also surprised us. It is the largest model and the only one trained specifically for Slovenian, but it hallucinated more often (66.7% per-answer rate) and refused fewer negatives (6/10) than the smaller multilingual baselines. Looking at individual questions explains the pattern: on the negative “*Recept za potico z orehi*”, GaMS produces a recipe rather than refusing, while Llama and Mistral correctly say the corpus has no such information. On the in-scope “*Kdaj poteka vpis v višji letnik*”, Llama answers verbatim from a single chunk (faithfulness 1.00) while GaMS adds details not present in any retrieved chunk (faithfulness 0.55, hallucination flagged). GaMS seems to prefer fluent Slovenian narration over strict context fidelity, which the judge penalises. We did not investigate whether this is prompt-format sensitivity, instruction-tuning data, or something else.

The human evaluation surfaces both successes and failures concretely. On Q43 (contact details), bge-m3 ($k_r=30$) retrieves all three programme e-mail addresses and phone numbers (9/10) while e5-base returns only one (3/10), confirming that a stronger embedder pays off on dense-fact queries. On Q27 (repeating the first year), all four RAG systems score 0/10 because the cross-encoder ranks a doctoral-programme regulation above the undergraduate rule; Claude, drawing on pre-training, answers 9/10. On Q61 (“Prevedi ta stavek v japonščino,” a request to translate into Japanese), the model produces an actual translation instead of refusing

(0/10) — more harmful than “I don’t know” because it looks plausible. The pattern across failures is consistent: wrong-domain retrieval is amplified by the generator, and missing context leads to hallucinated answers rather than refusals.

Direct comparison against prior university-RAG work [5, 6, 7] is difficult because their corpora and metrics differ (English admissions FAQs, accuracy on closed-set MCQs, etc.), so absolute numbers are not comparable. For this reason we limit ourselves to within-system comparisons (retrieval ablation, embedder sweep, generator sweep) and report all configurations under one consistent judge. The main caveat is the evaluation set size: 51 in-scope and 10 negative questions give limited statistical power, the language is mostly Slovenian, and even at $T=0$ a single LLM judge has per-question variability. A stratified manual review with Cohen’s κ against the Sonnet judge would be a natural next step. Extended failure analysis on a larger 147-question set and concrete improvement directions are provided in Appendices A and B.

Code and data availability

The complete pipeline, cleaned evaluation set, FAISS index, all judge/driver scripts, and SLURM launchers are released at the project repository (branch `eval-pipeline`). Seed URLs are in `raw_dataset/data_links.txt`. The `raw/` and `parsed.jsonl` directories are regenerable. We train no models. All embedders, the reranker, and the generators are open-weight models on Hugging Face Hub, and the judge uses Anthropic’s API. Only `ANTHROPIC_API_KEY` and `HF_TOKEN` are needed. `code/QUICKSTART.md` is the reproduction recipe.

Table 3. Human 0–10 evaluation on 60 questions (Q44 removed as duplicate; scorable count varies by configuration as some questions lacked a reference answer). **Claude (direct)** calls Sonnet with no retrieval, as a knowledge ceiling. All RAG variants use Llama-3.1-8B-Instruct and hybrid retrieval. k = chunks to the LLM; k_r = reranker candidate pool. * = default hyperparameters ($k=4$, $k_r=20$) from the automated sweep (Table 1). **Hall.** = answers scoring 0/10. Scores revised after reference-answer audit.

System	Emb.	k	k_r	Mean	≥ 7	≤ 2	Hall.
Claude direct (no RAG)	–	–	–	8.6	80%	3%	1
RAG (bge-m3, $k_r=30$)	bge-m3	6	30	5.1	41%	28%	10
RAG (bge-m3)*	bge-m3	4	20	4.8	40%	32%	11
RAG (e5-base)	e5-base	6	50	4.7	28%	23%	6
RAG (e5-base)*	e5-base	4	20	4.2	34%	40%	12

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1. Extended Manual Evaluation on the Full Development Set

To stress-test the final configuration we ran a second manual evaluation on the full 147-question development set (122 in-scope, 25 out-of-scope probes) using Llama-3.1-8B-Instruct with hybrid retrieval, $k=5$, $k_r=30$, $T=0$. Each answer was graded manually against the reference and the retrieved chunks.

Table 4. Extended manual evaluation

(Llama-3.1-8B-Instruct, hybrid + rerank, $k=5$, $k_r=30$, $T=0$). In-scope: $n=122$. Out-of-scope (OOS): $n=25$.

Partial: directionally correct but incomplete. **Incorrect:** factually wrong.

Category	Correct	Partial	Incorrect
In-scope ($n=122$)	97 (79.5%)	6 (4.9%)	19 (15.6%)
OOS probes ($n=25$)	10 (40%)	3 (12%)	12 (48%)

For in-scope questions the system answers correctly on 97/122 (79.5%), higher than the 5.1/10 mean in the smaller human evaluation because this pass uses binary grading and the larger set includes many straightforward factual look-ups. The out-of-scope refusal rate of 40% is lower than the automated-judge rate of 70% mainly because the extended probe set includes queries where retrieved chunks contain tangentially related text (investment funds, building floor plans, film screenings) that the model uses to construct a plausible-sounding off-topic answer instead of refusing.

The 19 in-scope errors fall into four patterns. **Wrong-domain retrieval** (9 cases): doctoral-programme regulations ranked above undergraduate ones (Q2, Q27), a 2010 annual report with the old address Tržaška cesta 25 instead of Večna pot 113 (Q104, Q126), FKKT lab hours instead of FRI library hours (Q8), or a master’s document for a bachelor question (Q80, Q84). **Misidentified intent** (4 cases): Q36 asks how long lab exercises remain valid but the model reports contact hours; Q114 confuses “students cannot be tutors” with “no tutoring for first-year students”; Q62 confuses home- and host-institution transcripts; Q23 confuses a confirmation email with the enrollment certificate. **Logical self-contradiction** (1 case): Q61 begins “No

payment needed” then immediately cites the regulation requiring payment for 4th/5th exam attempts. **Incomplete from poor retrieval** (5 cases): Q9 returns only two statuses from an Erasmus eligibility clause; Q14 names only the two co-run interdisciplinary doctorates, missing the primary Računalništvo in informatika programme; Q69 cites conflicting percentage thresholds without resolving the required credit points.

Out-of-scope failures confirm the same pattern: *baking bread at home* retrieves a food-science lab manual and produces oven advice; *Slovenian citizenship* retrieves an EU enrollment guide and answers about applying to university; *best programming language in 2025* retrieves Python course descriptions and recommends Python.

2. Possible Improvements

Several improvements follow directly from the failure analysis above. **Document-level metadata filtering.** Tag each chunk with its programme scope (*undergraduate*, *doctoral*, *general*) at index time and hard-filter candidates before the reranker; this would eliminate the entire class of wrong-domain retrievals (Q2, Q27, Q80, Q84) at no query-time cost. **Out-of-scope detection.** A lightweight query classifier or prompt-level guard would intercept off-topic requests before retrieval, preventing failures like Q61 (Japanese translation produced instead of refusal). **Surface-form disambiguation.** Adding document-type or section-header metadata to chunk representations would help the cross-encoder distinguish procedural pages from building-plan pages, reducing false positives from lexical overlap (e.g. “reception” matched to rental-premises documents in Q45). **LLM-based image parsing.** The OCR pass recovers text but discards figures and diagrams; applying systematic LLM-based extraction to image-bearing pages (as done manually for the building floor plan) would extend coverage to timetable graphics and organisational charts. **Larger evaluation set.** Expanding from 61 to 150–200 questions would give substantially more reliable mean scores and hallucination rate estimates.